



Sales Performance Analysis Report

Data Analytics Internship Project

Submitted By:

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Tools Used:

Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab

Dataset:

Superstore Sales Dataset

Project Objective:

The objective of this project is to analyze sales performance data to identify trends, profitable categories, regional performance, and seasonal sales patterns using data analytics and visualization techniques.

Executive Summary

The Sales Performance Analysis project was conducted to evaluate sales trends, profitability, regional performance, and customer demand patterns using the Superstore Sales dataset.

The analysis was performed using Python libraries such as Pandas, Matplotlib, and Seaborn within Google Colab. Important business metrics such as total sales, total profit, profit margin, and average order value were calculated to assess overall business performance.

Several visualizations were created to analyze monthly sales trends, category-wise sales contribution, regional performance, and product profitability.

The analysis revealed that certain categories and regions generated significantly higher revenue, while some products underperformed and affected profitability.

Overall, this project demonstrates how data analytics can support better business decision-making through actionable insights and visual analysis.

Dataset Overview

The dataset contains sales transaction records including:

- Product Categories
- Sub-Categories
- Sales
- Profit
- Quantity
- Regions
- Order Dates
- Customer Information

The dataset was cleaned and preprocessed before analysis to ensure accurate results.

KPI Analysis

The following KPIs were calculated:

- Total Sales
- Total Profit
- Profit Margin
- Average Order Value

These metrics provide an overview of the company's financial performance and operational efficiency.

▼ KPI 1 — Total Sales

```
[ ] total_sales = df['Sales'].sum()

    print("Total Sales:", total_sales)

▼ Total Sales: 2261536.7827000003
```

▼ KPI 2 — Total Orders

```
[ ] total_orders = df['Order_ID'].nunique()

    print("Total Orders:", total_orders)

▼ Total Orders: 4922
```

▼ KPI 3 — Total Customers

```
[ ] total_customers = df['Customer_ID'].nunique()

    print("Total Customers:", total_customers)

▼ Total Customers: 793
```

▼ KPI 4 — Average Order Value

Formula:

Average Order Value = $\frac{\text{Total Sales}}{\text{Number of Orders}}$

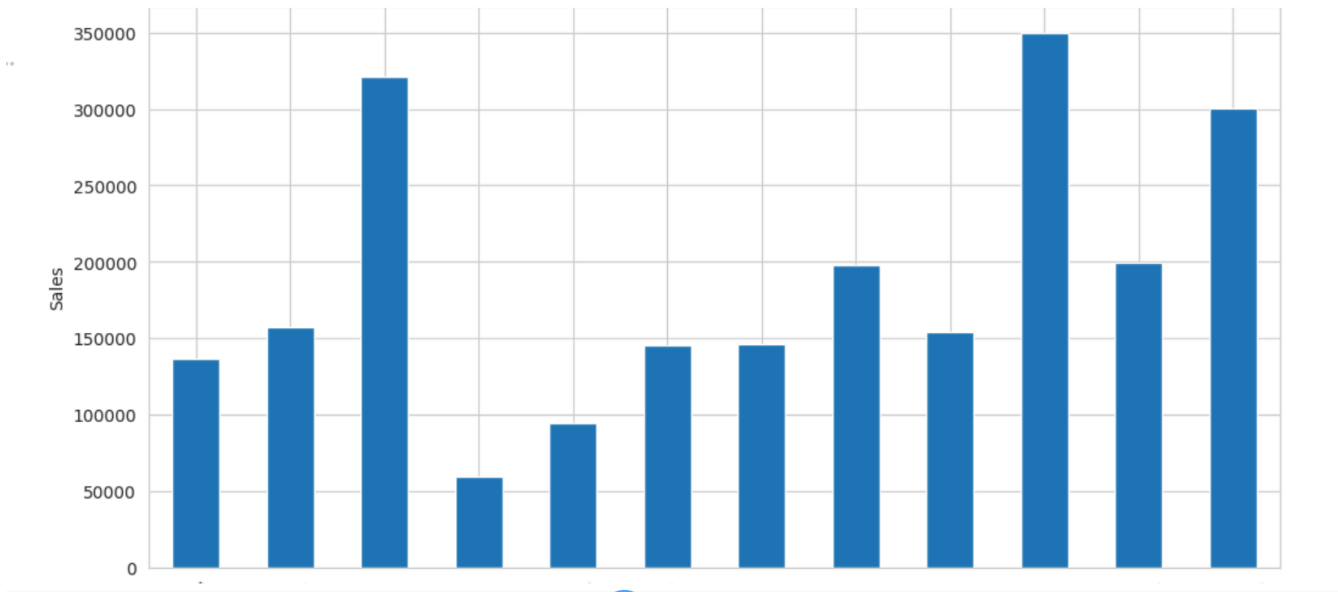
```
[ ] aov = df['Sales'].sum() / df['Order_ID'].nunique()

    print("Average Order Value:", round(aov,2))

▼ ... Average Order Value: 459.48
```

Monthly Sales Trend Analysis

The monthly sales trend analysis helps identify seasonal demand patterns and peak sales periods throughout the year.



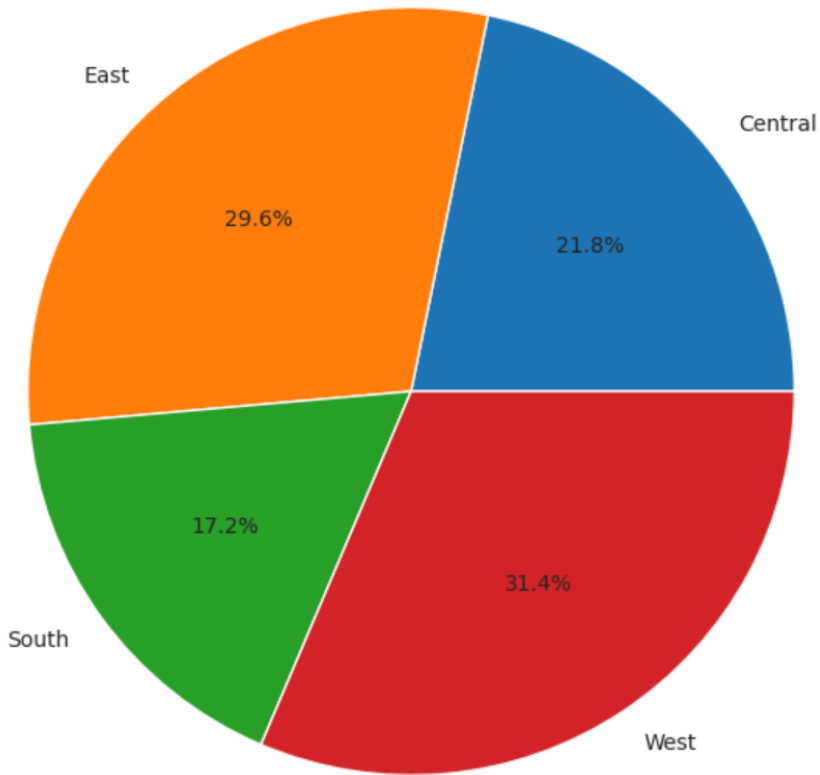
Insight

Sales fluctuate across different months, indicating seasonal purchasing behavior and variations in customer demand.

Regional Sales Analysis

Regional analysis compares sales performance across different regions to identify high-performing markets.

Sales by Region

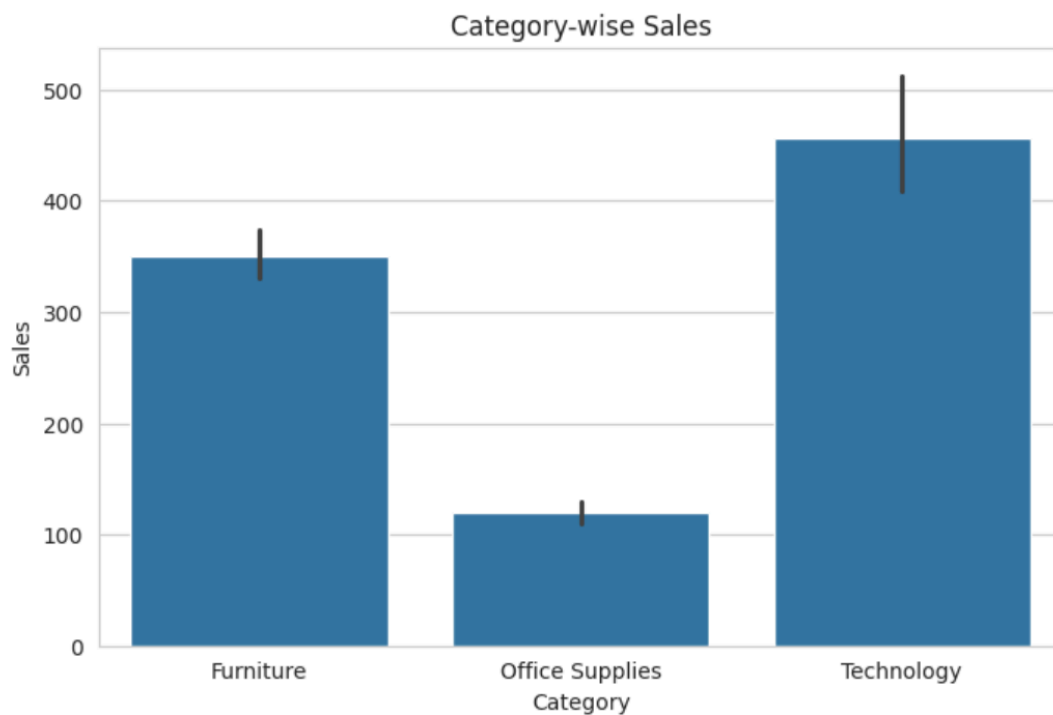


Insight

Some regions generated significantly higher sales revenue, indicating stronger customer demand and market performance.

Category-wise Sales Analysis

This analysis evaluates the contribution of different product categories toward overall sales performance.

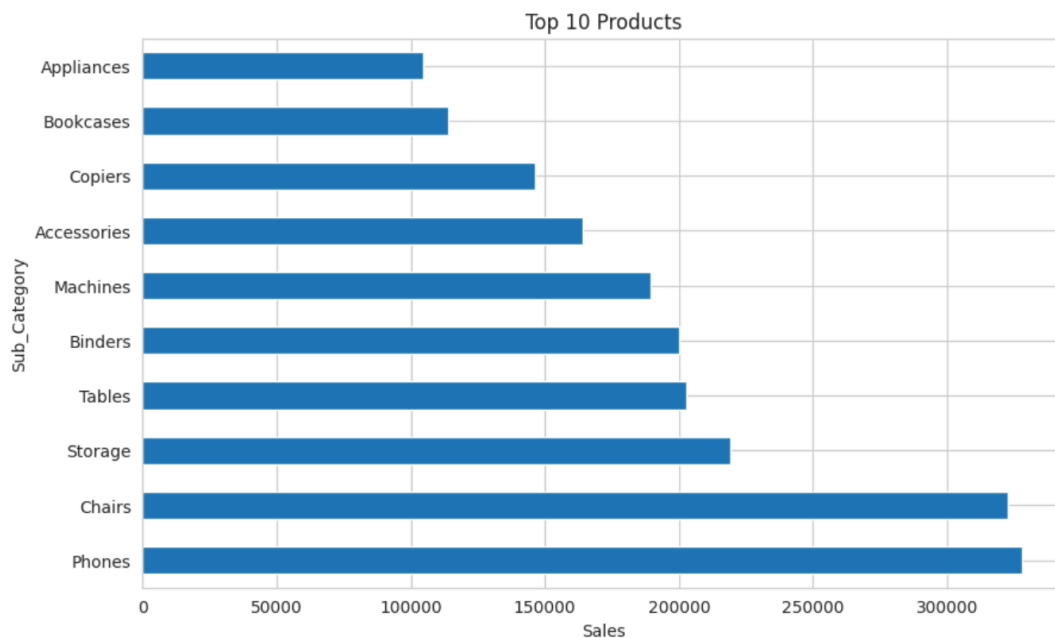


Insight

A few categories dominate total sales, reflecting higher customer preference and demand.

Top Performing Products

This section identifies products contributing the highest revenue to the business.

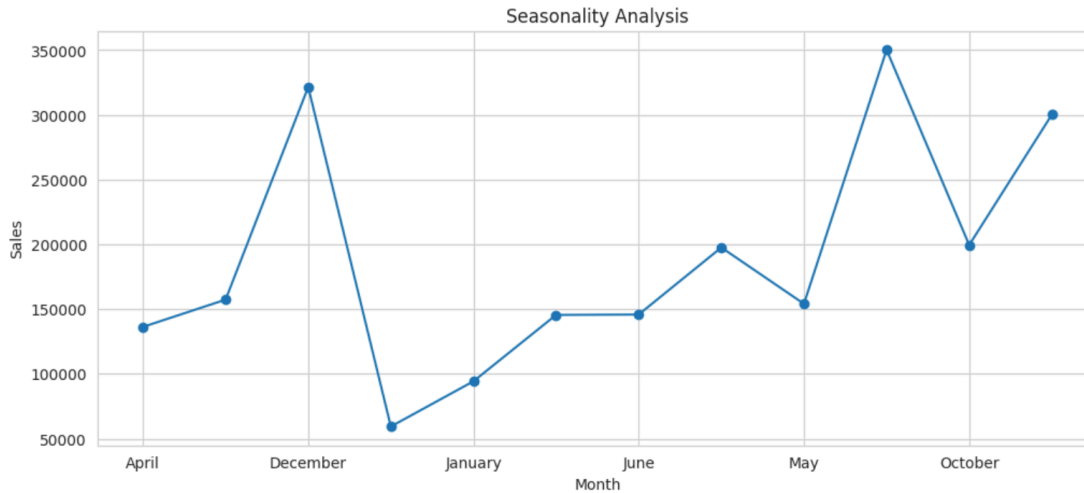


Insight

Top-performing products play a major role in revenue generation and should be prioritized in marketing and inventory planning.

Seasonal Sales Analysis

Seasonal analysis identifies recurring sales patterns across different months.



Insight

Sales vary across the year, reflecting seasonal customer purchasing behavior and demand changes.

Business Recommendations

1. Increase marketing efforts for top-performing products.
2. Improve sales strategies in low-performing regions.
3. Optimize inventory planning using seasonal demand patterns.
4. Focus more on high-profit categories.
5. Use data-driven decision-making to improve business growth and profitability.

Final Conclusion

The Sales Performance Analysis project successfully identified important business trends, profitable categories, regional performance differences, and seasonal demand patterns.

The project demonstrates how data analytics and visualization techniques can support strategic business decisions and improve overall operational efficiency.

